

[Product Information]

Ver.1.1

IMX455AQK-C

Diagonal 43.3 mm (Type 2.7) CMOS Image Sensor with Square Pixel for Color Cameras

Description

The IMX455AQK-C is a diagonal 43.3 mm (Type 2.7) CMOS active pixel type image sensor with a square pixel array and 61.17 M effective pixels. This sensor incorporates maximum 36 dB PGA circuit and 16-bit A/D converter. 16-bit digital output makes it possible to readout the signals of 61.17 M effective pixels at high-speed of 3.9 frame/s in still picture mode. In addition, this sensor can output 8K moving picture at 30 frame/s or 4K moving picture at 60 frame/s. This sensor is designed for use in consumer use digital still camera. When using this for another application, Sony Semiconductor Solutions Corporation does not guarantee the quality and reliability of this product. Therefore, don't use this for applications other than consumer use digital still camera.

In addition, individual specification change cannot be supported because this is a standard product.

Consult your Sony Semiconductor Solutions Corporation sales representative if you have any questions.

Features

- ◆ 35 mm full size CMOS active pixel type dots
- ◆ Input clock frequency 72 MHz
- ◆ Readout mode
 - All-pixel readout mode (+ digital overlap drive)
 - Various readout modes (*)
- ◆ Rolling shutter function moving picture mode
- ◆ H driver, V driver and serial communication circuit on chip
- ◆ +36 dB gain in CDS/PGA on chip (when A/D 16-bit, 14-bit, 12-bit)
- ◆ Built-in 11-bit/12-bit/14-bit/16-bit A/D converter
- ◆ 8 Lane SLVS-EC output
- ◆ R, G, B primary color mosaic filter on chip
- ◆ Back-illuminated type

* Please refer to the datasheet for binning/subsampling details of readout modes.

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Device Structure

◆ CMOS image sensor	
◆ Image size	Diagonal 43.3 mm (Type 2.7)
◆ Total number of pixels	9602 (H) × 6498 (V) approx. 62.39 M pixels
◆ Number of effective pixels	9576 (H) × 6388 (V) approx. 61.17 M pixels
◆ Number of active pixels	9568 (H) × 6380 (V) approx. 61.04 M pixels
◆ Chip size	40.961 mm (H) × 31.108 mm (V)
◆ Unit cell size	3.76 μm (H) × 3.76 μm (V)
◆ Optical black	Horizontal (H) direction: Front 16 pixels, Rear 0 pixel Vertical (V) direction: Front 22 pixels, Rear 0 pixel
◆ Package	294 pin LGA

Image Sensor Characteristics

(Tj = 60 °C)

Item		Value	Remarks
Sensitivity	Typ.	5630 LSB	1/30 s integration
Saturation signal	Min.	16383 LSB	Left for 0.4 s

Basic Drive Mode

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 0-a	9568 (H) × 6380 (V) approx. 61.04 pixels	3.97	16
Readout mode 0-b	9568 (H) × 6380 (V) approx. 61.04 pixels	9.98	14
Readout mode 0-c	9568 (H) × 6380 (V) approx. 61.04 pixels	17.91	12
Readout mode 0-e	9568 (H) × 6380 (V) approx. 61.04 pixels	21.33	10
Readout mode 0A-b	7696 (H) × 4320 (V) approx. 33.25 pixels	19.7	14
Readout mode 0A-c	7696 (H) × 4320 (V) approx. 33.25 pixels	30.99	12
Readout mode 0B-b	8208 (H) × 4320 (V) approx. 35.46 pixels	19.7	14
Readout mode 0B-c	8208 (H) × 4320 (V) approx. 35.46 pixels	30.65	12
Readout mode 0C-b	6288 (H) × 3528 (V) approx. 22.18 pixels	24.09	14
Readout mode 0C-c	6288 (H) × 3528 (V) approx. 22.18 pixels	37.89	12
Readout mode 3-a	9568 (H) × 6380 (V) approx. 61.04 pixels	1.98	16
Readout mode 3-b	9568 (H) × 6380 (V) approx. 61.04 pixels	4.98	14
Readout mode 3-c	9568 (H) × 6380 (V) approx. 61.04 pixels	8.95	12
Readout mode 3-e	9568 (H) × 6380 (V) approx. 61.04 pixels	10.66	10
Readout mode 6-c	4784 (H) × 3190 (V) approx. 15.26 pixels	38.45	12
Readout mode 6-d	4784 (H) × 3190 (V) approx. 15.26 pixels	45.33	12
Readout mode 6A-c	4784 (H) × 3190 (V) approx. 15.26 pixels	19.21	12
Readout mode 6A-d	4784 (H) × 3190 (V) approx. 15.26 pixels	22.65	12
Readout mode 6B-c	3848 (H) × 2160 (V) approx. 8.31 pixels	61.39	12
Readout mode 6C-c	3848 (H) × 2160 (V) approx. 8.31 pixels	30.67	12
Readout mode 7-c	4784 (H) × 3190 (V) approx. 15.26 pixels	38.43	12
Readout mode 7-d	4784 (H) × 3190 (V) approx. 15.26 pixels	45.3	12
Readout mode 7A-c	4784 (H) × 3190 (V) approx. 15.26 pixels	19.19	12

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 7A-d	4784 (H) × 3190 (V) approx. 15.26 pixels	22.62	12
Readout mode 7B-c	3848 (H) × 2160 (V) approx. 8.31 pixels	61.45	12
Readout mode 7C-c	3848 (H) × 2160 (V) approx. 8.31 pixels	30.67	12
Readout mode 8-c	3188 (H) × 2126 (V) approx. 6.78 pixels	57.52	12
Readout mode 8A-c	3188 (H) × 2126 (V) approx. 6.78 pixels	62.47	12
Readout mode 9-c	3188 (H) × 2124 (V) approx. 6.77 pixels	57.52	12
Readout mode 9A-c	3188 (H) × 2124 (V) approx. 6.77 pixels	62.47	12
Readout mode 12-c	3188 (H) × 2126 (V) approx. 6.78 pixels	28.7	12
Readout mode 12A-c	3188 (H) × 2126 (V) approx. 6.78 pixels	31.18	12
Readout mode 13-c	3188 (H) × 2124 (V) approx. 6.77 pixels	28.68	12
Readout mode 13A-c	3188 (H) × 2124 (V) approx. 6.77 pixels	31.15	12
Readout mode 14-c	3188 (H) × 1276 (V) approx. 4.07 pixels	95.34	12
Readout mode 14A-c	3188 (H) × 1276 (V) approx. 4.07 pixels	103.55	12
Readout mode 15-c	3188 (H) × 1274 (V) approx. 4.06 pixels	95.34	12
Readout mode 15A-c	3188 (H) × 1274 (V) approx. 4.06 pixels	103.55	12
Readout mode 18-c	3188 (H) × 1276 (V) approx. 4.07 pixels	47.52	12
Readout mode 18A-c	3188 (H) × 1276 (V) approx. 4.07 pixels	51.62	12
Readout mode 19-c	3188 (H) × 1274 (V) approx. 4.06 pixels	47.52	12
Readout mode 19A-c	3188 (H) × 1274 (V) approx. 4.06 pixels	51.62	12
Readout mode 20-d	3188 (H) × 910 (V) approx. 2.90 pixels	156.01	12
Readout mode 21-d	3188 (H) × 908 (V) approx. 2.89 pixels	156.01	12
Readout mode 22-d	3188 (H) × 708 (V) approx. 2.26 pixels	199.37	12
Readout mode 23-d	3188 (H) × 706 (V) approx. 2.25 pixels	199.37	12
Readout mode 24-d	3188 (H) × 424 (V) approx. 1.35 pixels	325.2	12
Readout mode 25-d	3188 (H) × 422 (V) approx. 1.35 pixels	325.2	12
Readout mode 26-d	3188 (H) × 236 (V) approx. 0.75 pixels	558.55	12
Readout mode 27-d	3188 (H) × 234 (V) approx. 0.75 pixels	558.55	12

[Product Information]

Ver.1.2

IMX455BQK-C

Diagonal 43.3 mm (Type 2.7) CMOS Image Sensor with Square Pixel for Color Cameras

Description

The IMX455BQK-C is a diagonal 43.3 mm (Type 2.7) CMOS active pixel type image sensor with a square pixel array and 61.17 M effective pixels. This sensor incorporates maximum 36 dB PGA circuit and 16-bit A/D converter. 16-bit digital output makes it possible to readout the signals of 61.17 M effective pixels at high-speed of 3.9 frame/s in still picture mode. In addition, this sensor can output 8K moving picture at 30 frame/s or 4K moving picture at 60 frame/s. This sensor is designed for use in consumer use digital still camera. When using this for another application, Sony Semiconductor Solutions Corporation does not guarantee the quality and reliability of this product. Therefore, don't use this for applications other than consumer use digital still camera.

In addition, individual specification change cannot be supported because this is a standard product. Consult your Sony Semiconductor Solutions Corporation sales representative if you have any questions.

Features

- ◆ 35 mm full size CMOS active pixel type dots
- ◆ Input clock frequency 72 MHz
- ◆ Readout mode
 - All-pixel readout mode (+ digital overlap drive)
 - Various readout modes (*)
- ◆ Rolling shutter function moving picture mode
- ◆ H driver, V driver and serial communication circuit on chip
- ◆ +36 dB gain in CDS/PGA on chip (when A/D 16-bit, 14-bit, 12-bit)
- ◆ Built-in 11-bit/12-bit/14-bit/16-bit A/D converter
- ◆ 8 Lane SLVS-EC output
- ◆ R, G, B primary color mosaic filter on chip
- ◆ Back-illuminated type
- ◆ Partially masked sensor pixels for Phase Detected Auto Focus (PDAF Pixel)

* Please refer to the datasheet for binning/subsampling details of readout modes.

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Device Structure

◆ CMOS image sensor	
◆ Image size Diagonal	43.3 mm (Type 2.7)
◆ Total number of pixels	9602 (H) × 6498 (V) approx. 62.39 M pixels
◆ Number of effective pixels	9576 (H) × 6388 (V) approx. 61.17 M pixels
◆ Number of active pixels	9568 (H) × 6380 (V) approx. 61.04 M pixels
◆ Chip size	40.961 mm (H) × 31.108 mm (V)
◆ Unit cell size	3.76 μm (H) × 3.76 μm (V)
◆ Optical black	Horizontal (H) direction: Front 16 pixels, Rear 0 pixel Vertical (V) direction: Front 22 pixels, Rear 0 pixel
◆ Package	294 pin LGA

Image Sensor Characteristics

(Tj = 60 °C)

Item		Value	Remarks
Sensitivity	Typ.	5080 LSB	1/30 s integration
Saturation signal	Min.	16383 LSB	Left for 0.4 s

Basic Drive Mode

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 0-a	9568 (H) × 6380 (V) approx. 61.04 pixels	3.97	16
Readout mode 0-b	9568 (H) × 6380 (V) approx. 61.04 pixels	9.98	14
Readout mode 0-c	9568 (H) × 6380 (V) approx. 61.04 pixels	17.91	12
Readout mode 0-e	9568 (H) × 6380 (V) approx. 61.04 pixels	21.33	10
Readout mode 3-a	9568 (H) × 6380 (V) approx. 61.04 pixels	1.98	16
Readout mode 3-b	9568 (H) × 6380 (V) approx. 61.04 pixels	4.98	14
Readout mode 3-c	9568 (H) × 6380 (V) approx. 61.04 pixels	8.95	12
Readout mode 3-e	9568 (H) × 6380 (V) approx. 61.04 pixels	10.66	10
Readout mode 6-c	4784 (H) × 3190 (V) approx. 15.26 pixels	38.45	12
Readout mode 6-d	4784 (H) × 3190 (V) approx. 15.26 pixels	45.33	12
Readout mode 6A-c	4784 (H) × 3190 (V) approx. 15.26 pixels	19.21	12
Readout mode 6A-d	4784 (H) × 3190 (V) approx. 15.26 pixels	22.65	12
Readout mode 7-c	4784 (H) × 3190 (V) approx. 15.26 pixels	38.43	12
Readout mode 7-d	4784 (H) × 3190 (V) approx. 15.26 pixels	45.30	12
Readout mode 7A-c	4784 (H) × 3190 (V) approx. 15.26 pixels	19.19	12
Readout mode 7A-d	4784 (H) × 3190 (V) approx. 15.26 pixels	22.62	12
Readout mode 8-c	3188 (H) × 2126 (V) approx. 6.78 pixels	57.52	12
Readout mode 9-c	3188 (H) × 2124 (V) approx. 6.77 pixels	57.52	12
Readout mode 12-c	3188 (H) × 2126 (V) approx. 6.78 pixels	28.70	12
Readout mode 13-c	3188 (H) × 2124 (V) approx. 6.77 pixels	28.68	12
Readout mode 14-c	3188 (H) × 1276 (V) approx. 4.07 pixels	95.34	12
Readout mode 15-c	3188 (H) × 1274 (V) approx. 4.06 pixels	95.34	12
Readout mode 18-c	3188 (H) × 1276 (V) approx. 4.07 pixels	47.52	12
Readout mode 19-c	3188 (H) × 1274 (V) approx. 4.06 pixels	47.52	12

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 20-d	3188 (H) × 910 (V) approx. 2.9 pixels	156.01	12
Readout mode 21-d	3188 (H) × 908 (V) approx. 2.89 pixels	156.01	12
Readout mode 22-d	3188 (H) × 708 (V) approx. 2.26 pixels	199.37	12
Readout mode 23-d	3188 (H) × 706 (V) approx. 2.25 pixels	199.37	12
Readout mode 24-d	3188 (H) × 424 (V) approx. 1.35 pixels	325.20	12
Readout mode 25-d	3188 (H) × 422 (V) approx. 1.35 pixels	325.20	12
Readout mode 26-d	3188 (H) × 236 (V) approx. 0.75 pixels	558.55	12
Readout mode 27-d	3188 (H) × 234 (V) approx. 0.75 pixels	558.55	12
Readout mode 29-c	3188 (H) × 2126 (V) approx. 6.78 pixels	55.76	12
Readout mode 35-c	3188 (H) × 1275 (V) approx. 4.06 pixels	90.61	12
Readout mode 41-d	3188 (H) × 911 (V) approx. 2.9 pixels	137.89	12
Readout mode 43-d	3188 (H) × 708 (V) approx. 2.26 pixels	172.88	12
Readout mode 45-d	3188 (H) × 424 (V) approx. 1.35 pixels	267.30	12
Readout mode 47-d	3188 (H) × 236 (V) approx. 0.75 pixels	418.66	12

[Product Information]

Ver.1.1

IMX455ALK-C

Diagonal 43.3 mm (Type 2.7) CMOS Image Sensor with Square Pixel for Monochrome Cameras

Description

The IMX455ALK-C is a diagonal 43.3 mm (Type 2.7) CMOS active pixel type image sensor with a square pixel array and 61.17 M effective pixels. This sensor incorporates maximum 36 dB PGA circuit and 16-bit A/D converter. 16-bit digital output makes it possible to readout the signals of 61.17 M effective pixels at high-speed of 3.9 frame/s in still picture mode. In addition, this sensor can output 8K moving picture at 30 frame/s or 4K moving picture at 60 frame/s.

This sensor is designed for use in consumer use digital still camera. When using this for another application, Sony Semiconductor Solutions Corporation does not guarantee the quality and reliability of this product. Therefore, don't use this for applications other than consumer use digital still camera.

In addition, individual specification change cannot be supported because this is a standard product.

Consult your Sony Semiconductor Solutions Corporation sales representative if you have any questions.

Features

- ◆ 35 mm full size CMOS active pixel type dots
- ◆ Input clock frequency 72 MHz
- ◆ Readout mode
 - All-pixel readout mode (+ digital overlap drive)
 - Various readout modes (*)
- ◆ Rolling shutter function moving picture mode
- ◆ H driver, V driver and serial communication circuit on chip
- ◆ +36 dB gain in CDS/PGA on chip (when A/D 16-bit, 14-bit, 12-bit)
- ◆ Built-in 11-bit/12-bit/14-bit/16-bit A/D converter
- ◆ 8 Lane SLVS-EC output
- ◆ Back-illuminated type

* Please refer to the datasheet for binning/subsampling details of readout modes.

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Device Structure

- ◆ CMOS image sensor
- ◆ Image size Diagonal 43.3 mm (Type 2.7)
- ◆ Total number of pixels 9602 (H) × 6498 (V) approx. 62.39 M pixels
- ◆ Number of effective pixels 9576 (H) × 6388 (V) approx. 61.17 M pixels
- ◆ Number of active pixels 9568 (H) × 6380 (V) approx. 61.04 M pixels
- ◆ Chip size 40.961 mm (H) × 31.108 mm (V)
- ◆ Unit cell size 3.76 μm (H) × 3.76 μm (V)
- ◆ Optical black Horizontal (H) direction: Front 16 pixels, Rear 0 pixel
Vertical (V) direction: Front 22 pixels, Rear 0 pixel
- ◆ Package 294 pin LGA

Image Sensor Characteristics

(Tj = 60 °C)

Item		Value	Remarks
Sensitivity	Typ.	5060 LSB	1/30 s integration
Saturation signal	Min.	16383 LSB	Left for 0.4 s

Basic Drive Mode

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 0-a	9568 (H) × 6380 (V) approx. 61.04 pixels	3.97	16
Readout mode 0-b	9568 (H) × 6380 (V) approx. 61.04 pixels	9.98	14
Readout mode 0-c	9568 (H) × 6380 (V) approx. 61.04 pixels	17.91	12
Readout mode 0-e	9568 (H) × 6380 (V) approx. 61.04 pixels	21.33	10
Readout mode 0A-b	7696 (H) × 4320 (V) approx. 33.25 pixels	19.7	14
Readout mode 0A-c	7696 (H) × 4320 (V) approx. 33.25 pixels	30.99	12
Readout mode 0B-b	8208 (H) × 4320 (V) approx. 35.46 pixels	19.7	14
Readout mode 0B-c	8208 (H) × 4320 (V) approx. 35.46 pixels	30.65	12
Readout mode 0C-b	6288 (H) × 3528 (V) approx. 22.18 pixels	24.09	14
Readout mode 0C-c	6288 (H) × 3528 (V) approx. 22.18 pixels	37.89	12
Readout mode 3-a	9568 (H) × 6380 (V) approx. 61.04 pixels	1.98	16
Readout mode 3-b	9568 (H) × 6380 (V) approx. 61.04 pixels	4.98	14
Readout mode 3-c	9568 (H) × 6380 (V) approx. 61.04 pixels	8.95	12
Readout mode 3-e	9568 (H) × 6380 (V) approx. 61.04 pixels	10.66	10
Readout mode 6-c	4784 (H) × 3190 (V) approx. 15.26 pixels	38.45	12
Readout mode 6-d	4784 (H) × 3190 (V) approx. 15.26 pixels	45.33	12
Readout mode 6A-c	4784 (H) × 3190 (V) approx. 15.26 pixels	19.21	12
Readout mode 6A-d	4784 (H) × 3190 (V) approx. 15.26 pixels	22.65	12
Readout mode 6B-c	3848 (H) × 2160 (V) approx. 8.31 pixels	61.39	12
Readout mode 6C-c	3848 (H) × 2160 (V) approx. 8.31 pixels	30.67	12
Readout mode 7-c	4784 (H) × 3190 (V) approx. 15.26 pixels	38.43	12
Readout mode 7-d	4784 (H) × 3190 (V) approx. 15.26 pixels	45.3	12
Readout mode 7A-c	4784 (H) × 3190 (V) approx. 15.26 pixels	19.19	12

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 7A-d	4784 (H) × 3190 (V) approx. 15.26 pixels	22.62	12
Readout mode 7B-c	3848 (H) × 2160 (V) approx. 8.31 pixels	61.45	12
Readout mode 7C-c	3848 (H) × 2160 (V) approx. 8.31 pixels	30.67	12
Readout mode 8-c	3188 (H) × 2126 (V) approx. 6.78 pixels	57.52	12
Readout mode 8A-c	3188 (H) × 2126 (V) approx. 6.78 pixels	62.47	12
Readout mode 9-c	3188 (H) × 2124 (V) approx. 6.77 pixels	57.52	12
Readout mode 9A-c	3188 (H) × 2124 (V) approx. 6.77 pixels	62.47	12
Readout mode 12-c	3188 (H) × 2126 (V) approx. 6.78 pixels	28.7	12
Readout mode 12A-c	3188 (H) × 2126 (V) approx. 6.78 pixels	31.18	12
Readout mode 13-c	3188 (H) × 2124 (V) approx. 6.77 pixels	28.68	12
Readout mode 13A-c	3188 (H) × 2124 (V) approx. 6.77 pixels	31.15	12
Readout mode 14-c	3188 (H) × 1276 (V) approx. 4.07 pixels	95.34	12
Readout mode 14A-c	3188 (H) × 1276 (V) approx. 4.07 pixels	103.55	12
Readout mode 15-c	3188 (H) × 1274 (V) approx. 4.06 pixels	95.34	12
Readout mode 15A-c	3188 (H) × 1274 (V) approx. 4.06 pixels	103.55	12
Readout mode 18-c	3188 (H) × 1276 (V) approx. 4.07 pixels	47.52	12
Readout mode 18A-c	3188 (H) × 1276 (V) approx. 4.07 pixels	51.62	12
Readout mode 19-c	3188 (H) × 1274 (V) approx. 4.06 pixels	47.52	12
Readout mode 19A-c	3188 (H) × 1274 (V) approx. 4.06 pixels	51.62	12
Readout mode 20-d	3188 (H) × 910 (V) approx. 2.90 pixels	156.01	12
Readout mode 21-d	3188 (H) × 908 (V) approx. 2.89 pixels	156.01	12
Readout mode 22-d	3188 (H) × 708 (V) approx. 2.26 pixels	199.37	12
Readout mode 23-d	3188 (H) × 706 (V) approx. 2.25 pixels	199.37	12
Readout mode 24-d	3188 (H) × 424 (V) approx. 1.35 pixels	325.2	12
Readout mode 25-d	3188 (H) × 422 (V) approx. 1.35 pixels	325.2	12
Readout mode 26-d	3188 (H) × 236 (V) approx. 0.75 pixels	558.55	12
Readout mode 27-d	3188 (H) × 234 (V) approx. 0.75 pixels	558.55	12